Table 8.1 Estimated rates of convergence, perturbation growth, and efficiency for different methods

Thermostat	Convergence rate	Error (to lowest order in t)	Efficiency
Langevin	γ	γt	≈ 1
Velocity Rescaling	$4\,\frac{\mathrm{Av}_\beta\{K^2\Phi(K)\}}{\mathrm{TN_d k_B}}$	$\frac{2}{\mathrm{N_d}}\mathrm{Av}_\beta\{K\Phi(K)\}\,t$	$O(\mathrm{N_d})$
VR with $\Phi(K)=\frac{\gamma_{\mathrm{VR}}}{2K}$	γ_{VR}	$\frac{\gamma_{\mathrm{VR}}}{\mathrm{N_d}}t$	$\approx \mathrm{N_d}$
NHL	$\sqrt{\frac{\mathrm{N_d}k_\mathrm{B}\mathrm{T}}{\mu}}$ (at critical damping)	$\frac{k_\mathrm{B}\mathrm{T}}{2\mu}t^2$	$\approx 2\mathrm{N_d}$

8.5 Non-Newtonian Dynamics-Based Iterations for Molecular Sampling

Given that we can decompose the stochastic dynamics into deterministic part and stochastic process, and approximately or exactly solve the parts, it is common to iterate the maps for each building block sequentially in such a way that the calculation approximates the dynamics of the system. This is, very likely, an unnecessary restriction on the sampling strategy. The crucial things that effect the ultimate success of the method are that the parts themselves preserve the measure (or approximately so), the stages are volume preserving, and that the stochastic processes are sufficiently strongly coupled so that the overall iteration is ergodic.

There is essentially no need to use maps based on Newtonian deterministic dynamics. There are many alternative dynamics models which leave invariant the canonical measure, going well beyond those considered so far. To illustrate this with one intriguing example, we explore here the stochastic dynamical system in $2\mathrm{N_c}+1$ variables defined by

$$\mathrm{d}\boldsymbol{q} = \xi M^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.33}$$

$$\mathrm{d}\boldsymbol{p} = -\gamma\boldsymbol{p}\mathrm{d}t + \sigma\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}, \tag{8.34}$$

$$\mathrm{d}\xi = -[(\boldsymbol{M}^{-1}\boldsymbol{p})\cdot\nabla U(\boldsymbol{q})/\mu + \gamma'\xi]\mathrm{d}t + \sigma'\mathrm{d}Z, \tag{8.35}$$

where $\mathbf{W}$ is a vector of $\mathrm{N_c}$ Wiener processes, Z is a scalar Wiener process and $\sigma = \sqrt{2\mathrm{k_B T}\gamma}$, and $\sigma' = \sqrt{2\mathrm{k_B T}\gamma'/\mu}$ this system preserves the augmented distribution with density

$$\tilde{\rho}(\boldsymbol{q},\boldsymbol{p},\xi) = e^{-\beta(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2+U(\boldsymbol{q})+\mu\xi^2/2)}.$$

For this system, it is possible to show that the Hörmander condition holds. Setting all parameters to 1 and taking $\boldsymbol{M}=\boldsymbol{I}$ (without loss of generality in this demonstration) we write our SDE in the form $\mathrm{d}\varphi = D + \sum_{i=1}^{\mathrm{N_c}+1} Y_i \mathrm{dW}_i$, where φ is the vector of all phase variables, $Y_i = \boldsymbol{e}_{\mathrm{N_c}+i}$, for $i = 1,\ldots,\mathrm{N_c}+1$, is the $i+\mathrm{N_c}$th basis function in

$\mathbb{R}^{2N_c+1}$, the W_i are independent Wiener processes, and D is the "deterministic part:"

$$D = \begin{bmatrix} \xi \boldsymbol{p} \\ -\boldsymbol{p} \\ -\boldsymbol{p} \cdot \nabla U - \xi \end{bmatrix}.$$

Then we calculate directly

$$D' = [D, Y_{N_c+1}] = \begin{bmatrix} \boldsymbol{p} \\ 0 \\ 1 \end{bmatrix},$$

and from this we see that $[D', Y_i] = \boldsymbol{e}_i$ for $i = 1, 2, \ldots, N_c$. Thus all the basis vectors of $\mathbb{R}^{2N_c+1}$ are found in the Lie algebra of the components of the Hörmander decomposition; as we know, this is the key step in the proof of ergodicity. Thus the method (8.33)–(8.35) is an alternative for molecular sampling dynamics, even though it is not based on a Newtonian model. Leaving the stochastic term off the final equation (i.e. in ξ) is easily shown to destroy the ergodicity property (the Hörmander condition is then violated).

Now consider the splitting into deterministic and OU parts. The deterministic part

$$\dot{\boldsymbol{q}} = \xi M^{-1}\boldsymbol{p}, \quad \dot{\boldsymbol{p}} = 0, \quad \dot{\xi} = -(\boldsymbol{M}^{-1}\boldsymbol{p}) \cdot \nabla U(\boldsymbol{q})/\mu, \tag{8.36}$$

preserves volume and the function $U(\boldsymbol{q}) + \mu\xi^2/2$ and hence also the distribution with density $\tilde{\rho}_A(\boldsymbol{q}, \xi) = e^{-\beta(U(\boldsymbol{q})+\mu\xi^2/2)}$. By the same token the OU processes collectively preserve $\tilde{\rho}_B(\boldsymbol{p}, \xi) = e^{-\beta(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2+\mu\xi^2/2)}$. In combination the full dynamics preserves the extended canonical measure. If we iterate the two parts (assuming the deterministic part is solved exactly) the distribution will converge to the exact canonical distribution, due to especially to incorporation of $N_c + 1$ noise processes.

The OU process defining the fluctuations in $\boldsymbol{p}$ is completely decoupled from $\boldsymbol{q}$ and ξ, so it is perhaps most natural to think of the iteration as a random walk on the reduced space $(\boldsymbol{q}, \xi)$ where the directions are generated by an auxiliary OU process. Then the algorithm proceeds as follows: (1) choose a random direction $\boldsymbol{p}$ using the OU process, and (2) sample the reduced (stochastic) dynamics for $(\boldsymbol{q}, \xi)$ in the direction $\boldsymbol{p}$. We refer to this procedure as *stochastic line sampling* (SLS), for the following reason. With $\boldsymbol{p}$ fixed in step (2), the motion of $\boldsymbol{q}$ is confined to the straight line: $\boldsymbol{q} = \boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p}$, where η is a scalar function of time. Inserting this coordinate change in (8.36) we see that the equations reduce to $\dot{\eta} = \xi$, $\dot{\xi} = -\partial/\partial\eta U(\boldsymbol{q}_0 + \eta\boldsymbol{M}^{-1}\boldsymbol{p})$. This is a one-degree of freedom system and hence is theoretically exactly solvable in quadratures, although the complexity of the force will render this impractical in calculations, and, as mentioned above, an additional stochastic process must be incorporated to drive ξ.